Cryptography & Network Security

Classical Encryption: Substitution, Transposition, Caesar & Shift Ciphers

1 Classical Encryption Techniques

Plaintext sent in the open over a channel invites attacks: **passive** (eavesdropping, reading the data) and **active** (altering or injecting data). Encryption turns plaintext into ciphertext so an interceptor learns nothing.

Classical techniques are **obsolete**: easy to break, keys recovered quickly. We study them because modern ciphers are built on the same two ideas: *substitution* and *transposition*.

1.1 Substitution technique

Letters are **replaced** by other letters or symbols. Positions stay the same.

Key (a mapping table): a→M, b→X, g→A, x→Z

Plaintext	b	a	g
	↓	↓	↓
Ciphertext	**X**	**M**	**A**

- The mapping table *is* the key.
- Sender and receiver share the same table ⇒ symmetric encryption.

1.2 Transposition technique

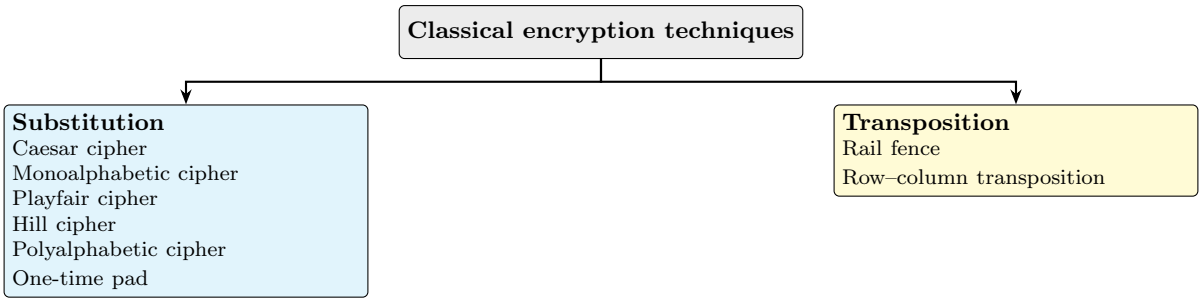
Apply a **permutation**: the same letters are **rearranged**. No new letters appear.

Plaintext `NESO` → `ESON`, `SONE`, `ONES`, `ENOS`, ... e.g. `ESON` takes letters in order $(2, 3, 4, 1)$; `SONE` takes $(3, 4, 1, 2)$.

A 4-letter word has only $4! = 24$ arrangements, so short messages are easy to guess. Transposition is only effective for **long** plaintexts ($n!$ grows fast).

1.3 Substitution vs. transposition

	Substitution	**Transposition**
What changes	The letters themselves	The order of the letters
Positions	Unchanged	Permuted
Letters in ciphertext	New letters / symbols	Exactly the plaintext letters
Letter frequencies	Moved onto different symbols	Identical to plaintext



2 Caesar Cipher

Earliest and simplest substitution cipher, used by Julius Caesar. Replace each letter with the letter **three places further down** the alphabet, wrapping from `z` back to `a`.

2.1 Letters as numbers (start at 0)

A	B	C	D	E	F	G	H	I	J	K	L	M
0	1	2	3	4	5	6	7	8	9	10	11	12
N	O	P	Q	R	S	T	U	V	W	X	Y	Z
13	14	15	16	17	18	19	20	21	22	23	24	25

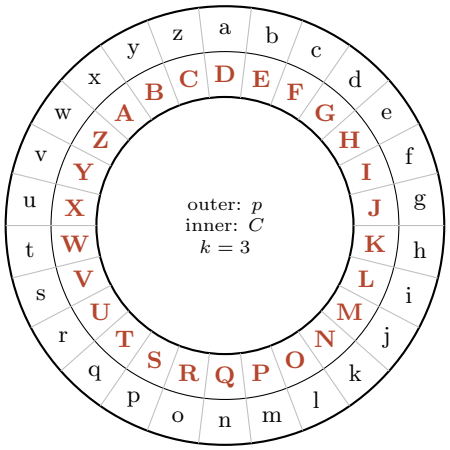
Numbering from 0 matches the remainders mod 26, which are exactly $0, 1, \ldots, 25$.

2.2 Algorithm

Encryption and decryption

$$C = E(p, k) = (p + k) \bmod 26$$
$$p = D(C, k) = (C - k) \bmod 26$$

Caesar cipher: $k = 3$.



- p = plaintext letter's number, C = ciphertext letter's number, k = key (shift).
- Convention: plaintext in lowercase, ciphertext in UPPERCASE.
- **mod gives the remainder, not the quotient.** 27 mod 26 = 1.
- Negative during decryption? Add 26. $-2 \bmod 26 = 24$.

2.3 Worked example: encrypt "neso academy" ($k = 3$)

Plaintext	n	e	s	o	a	c	a	d	e	m	y
p	13	4	18	14	0	2	0	3	4	12	24
$p+3$	16	7	21	17	3	5	3	6	7	15	**27**
mod26	16	7	21	17	3	5	3	6	7	15	**1**
Ciphertext	**Q**	**H**	**V**	**R**	**D**	**F**	**D**	**G**	**H**	**P**	**B**

Answer: `QHVR DFDGHPB`. The only wrap-around is y: $24 + 3 = 27$, and $27 = 26 \times 1 + \mathbf{1}$, so $C = 1 =$ `B`. (Counting on: y → z → a → **b**.) For the other letters you can just count three letters forward.

2.4 Homework 1: decrypt "QHWZRUN VHFXULWB" ($k = 3$)

Ciphertext	Q	H	W	Z	R	U	N	V	H	F	X	U	L	W	B
C	16	7	22	25	17	20	13	21	7	5	23	20	11	22	1
$C-3$	13	4	19	22	14	17	10	18	4	2	20	17	8	19	-2
mod26	13	4	19	22	14	17	10	18	4	2	20	17	8	19	**24**
Plaintext	n	e	t	w	o	r	k	s	e	c	u	r	i	t	y

Answer: **network security**. Note the last letter: $1 - 3 = -2$, add 26 ⇒ 24 = y.

3 Shift Cipher (the general form)

Same formulas, but the key k can be any of $1, 2, \ldots, 25$. **Caesar cipher = shift cipher with $k = 3$.**

- $k = 0$ is useless: $C = (p + 0) \bmod 26 = p$, so the ciphertext equals the plaintext. (Homework: "Neso", $k = 0$ ⇒ `NESO`.)
- $k = 26$ is the same as $k = 0$. So there are only **25 usable keys**.

Example: "Neso", $k = 4$: N 13+4=17 →**R**, e 4+4=8 →**I**, s 18+4=22 →**W**, o 14+4=18 →**S** ⇒ `RIWS`.

Homework 2: "cryptography", $k = 4$:

Plaintext	c	r	y	p	t	o	g	r	a	p	h	y
p	2	17	24	15	19	14	6	17	0	15	7	24
$p+4 \bmod 26$	6	21	**2**	19	23	18	10	21	4	19	11	**2**
Ciphertext	G	V	C	T	X	S	K	V	E	T	L	C

Answer: `GVCTXSKVETLC` (both y's wrap: $24 + 4 = 28 \equiv 2$).

In code: the 0–25 numbering is a convention. With ASCII (`A`=65, `a`=97), shift to 0 first, then back: $C = \big((p - 65 + k) \bmod 26\big) + 65$ for uppercase (use 97 for lowercase).

3.1 Pros and cons of the Caesar cipher

Pros

1. Simple.
2. Easy to implement.

Cons

1. The encryption and decryption algorithms are known.
2. Only 25 keys to try ⇒ vulnerable to **brute-force attack**.
3. The plaintext language is known and easily recognisable, so you can tell when you have hit the right key.

4 Brute-Force Attack on the Caesar/Shift Cipher

Brute force: try every possible key until readable plaintext appears. With 25 keys this takes seconds by hand.

Ciphertext: `SQDYMZK`. Shift every letter forward by 0, 1, 2, . . . , 25 and look for a real word:

Fwd	Back	Result
0	26	SQDYMZK
1	25	TREZNAL
2	24	USFAOBM
3	23	VTGBPCN
4	22	WUHCQDO
5	21	XVIDREP
6	20	YWJESFQ
7	19	ZXKFTGR
8	18	AYLGUHS
9	17	BZMHVIT
10	16	CANIWJU
11	15	DBOJXKV*
12	14	ECPKYLW
13	13	FDQLZMX
14	12	GERMANY
15	11	HFSNBOZ
16	10	IGTOCPA
17	9	JHUPDQB
18	8	KIVQERC
19	7	LJWRFSD
20	6	MKXSGTE
21	5	NLYTHUF
22	4	OMZUIVG*
23	3	PNAVJWH
24	2	QOBWKXI
25	1	RPCXLYJ

* Recomputed; the notes had these rows slightly wrong (see corrections box).

- Forward + back = 26, because shifting forward 14 is the same as shifting back 12: $C + 14 \equiv C - 12 \pmod{26}$.
- Only one row is English: **GERMANY**. The attacker now has the plaintext *and* the key.
- **Which number is the key?** The sender encrypted with $k = \mathbf{12}$ (G $6+12 = 18 =$ S, . . . , Y $24+12 = 36 \equiv 10 =$ K). The 14 is the forward shift that *undoes* it. The lecture calls 14 "the key"; be clear which one you mean in an exam.

Decryption vs. cryptanalysis

Decryption: the receiver *has* the key and reverses the encryption.
Cryptanalysis: an attacker *without* the key recovers the plaintext and/or the key (e.g. by brute force).

Exam tips

- Substitution = replace letters; transposition = rearrange letters. Know the list of ciphers under each.
- Write both formulas: $C = (p + k) \bmod 26$, $p = (C - k) \bmod 26$. Caesar is $k = 3$.
- Always show the wrap-around step (e.g. $27 \bmod 26 = 1$) and fix negatives by adding 26.
- Key space is 25 (not 26): $k = 0$ changes nothing.
- Three cons of Caesar: algorithms known, only 25 keys, language recognisable.
- For brute force: draw the table, box the readable row, state the key.

Corrections to the notes / slides

- "neso academy" encrypts to `QHVR DFDGHPB`. The notes' final answer `QHVRDFDGHB` drops the `P`.
- Brute-force table: row 11 is `DBOJXKV` and row 22 is `OMZUIVG`. Row 12 (`ECPKYLW`) was already correct.
- The notes call con #1 a "Kerckhoffs violation". It isn't: public algorithms are normal (see the intro sheet). The problem is a public algorithm *plus* only 25 keys.
- Slide typo: "used *be* Julius Caesar" → "used *by*".